A Simple Model for Capture and Emission Time Constants of Random Telegraph Signal Noise

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***Abstract*—A simple model for electron capture and emission process in oxide traps is introduced for analysis of random telegraph signal noise. Multiphonon emission capture theory is used for a "more fundamental" thermally activated capture cross-sectional model. Basically, Shockley–Read–Hall statistics is considered for time constant modeling. Particularly, the thermally activation model with the quantum effect is applied to emission time constant modeling. It is shown that the capture and emission process are controlled by characteristics of traps. Especially, the physical meaning of the lattice relaxation energy which is the most important model parameter and the trap characteristic is clarified.**

***Index Terms*—Capture, capture cross section, emission, noise, oxide trap, random telegraph signal (RTS), time constant modeling.**

I. Introduction

As CMOS devices are down-scaled, the effect of an individual defect on device performance becomes an important issue. In submicron MOSFET, the random telegraph signal (RTS) noise, characterized by the discrete switching performance of the channel current, is observed through the capture and emission of conduction carriers in the individual interfacial defect of gate dielectrics [1]–[3]. If the trap energy level is within a few $k_B T$ of the electron Fermi level where k_B is Boltzmann's constant and T is the absolute temperature, a trapping phenomenon takes place [4]. Thus, the RTS noise provides information of individual traps.

A conventional method for the capture process analysis, however, is assumed to be controlled by the tunneling process [5]. This assumption can be confronted with problem. The capture time constant extracted from the tunneling time restriction set by the dielectric thickness is orders of magnitude larger than the expected time [6]. This means that the capture process not only occurs solely by the tunneling mechanism.

In a previous report, a simple model of multiphonon emission (MPE) capture cross sections that is only valid at high temperatures was verified [7]. In this study, a "more fundamental" model for capture cross-sectional determination, originating from a simple theory of MPE capture, is used, and is shown to be valid at room temperature. For the emission process analysis, the thermally activation model with the quantum effect and Shockley–Read–Hall (SRH) statistics are considered. Finally, it is shown that the capture and emission process is controlled by trap characteristics such as relaxation energy.

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II. Capture and Emission Time Constant Model

A. Capture Time Constant Model

Generally, the capture process between conduction carriers and an oxide trap is governed by SRH statistics [8], as given by

$$\tau_c = \frac{1}{\sigma_C \langle \upsilon \rangle n}. \tag{1}$$

Here, σ_C is the capture cross section, $\langle \upsilon \rangle$ is the mean thermal velocity, and n is electron density.

Previously, the MPE model was applied to a capture time model by Palma *et al*. [9]. In their model, however, the MPE theory was adopted for an electron tunnel transition mechanism between a trap site and a channel, and was used to calculate total capture probability. That application method differs from the one described in this paper. For simplification of the model application process, we use MPE theory for the determination of σ_C. The concise and fundamental model for thermally activated σ_C with consideration of the MPE theory was derived by Henry and Lang [10] and is given by

$$\sigma_C = \sigma_{\rm oc} \exp\left[-\frac{E_{\rm oc}}{k_B T}\right]. \tag{2}$$

With reference to Fig. 1, $E_{\rm oc}$ is a function of the activation energy E_B, and is given by $E_{\rm oc} = E_B - k_B T$. $\sigma_{\rm oc}$ is carried out for instances of capture by an attractive neutral impurity, and is given by

$$\sigma_{\rm oc} = \sqrt{\frac{E_1}{S\hbar\omega}} \left(\frac{\pi^2}{2e}\right) \left(\frac{\hbar^2}{2m^* k_B T}\right) \tag{3}$$

where $\hbar$ is a reduced Planck constant, m^* is the effective mass, and e is Euler's number. Typically, $E_1 \approx 0.06$ eV [10]. The parameter S, proposed by Huang and Rhys [11], is related to the lattice relaxation energy $E_{\rm relax}$, and given by $S = E_{\rm relax}/\hbar\omega$, where $\hbar\omega$ is the characteristic phonon energy. This approach is similar to that of Kirton and Uren [12] in appearance because the form of the σ_C model is almost the same. The σ_C model

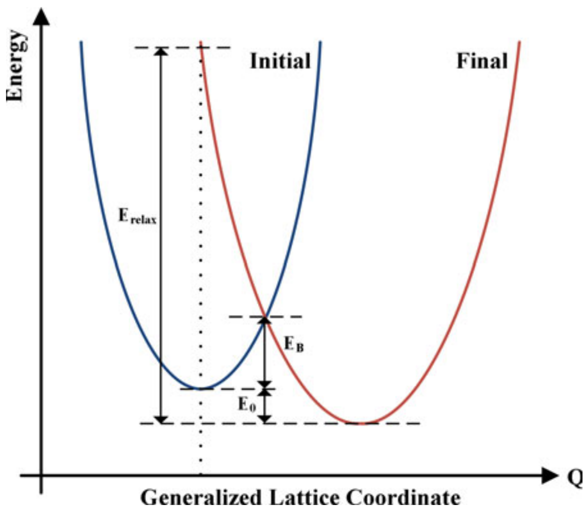

Fig. 1. Configuration coordinate diagrams of energy versus the generalized lattice coordinate Q. E_0 is the total binding energy.

used by Kirton and Uren is given by

$$\sigma_{C,\mathrm{KU}} = \sigma_o \exp\left[-\frac{E_B}{k_B T}\right] \tag{4}$$

where σ_o is a cross-sectional prefactor proposed by Schulz and Johnson [13]. That model and a constant value of σ_o are valid at high temperatures [13]. This is mainly considered for including thermal effect and caused by so many assumptions and simplifications. However, variation in σ_C can be associated with changes in either temperature or energy [14]. Thus, the effects of trap energy need to be also taken into account for low-temperature region. Thus, the more fundamental energy-based models represented by (2) and (3) are expected to be accurate at low temperatures.

B. Emission Time Constant

The emission time constant model based on SRH statistics is given by

$$\tau_{e\mathrm{SRH}} = \frac{\tau_c}{g} \exp\left[-\frac{E_T - E_F}{k_B T}\right] \tag{5}$$

where g is the degeneracy factor. Generally, $g = 1$ for the case of RTS noise analysis. E_T and E_F are the trap-level and Fermi-level energies, respectively, and $\tau_{e\mathrm{SRH}}$ is the time that the trapped carrier is emitted when using two steps. As shown in path 1 in Fig. 2, the first step is from the trap to an interfacial state near the Fermi level of the surface and the second step is from the interfacial state to the conduction band. Another emission path, however, is one in which the trapped carrier emits to the conduction band directly, see path 2 in Fig. 2. For calculation of this, the fundamental thermally activated emission time model is used, as given by

$$\tau_{eT} = \frac{g}{\sigma_C \langle \upsilon \rangle N} \exp\left[\frac{\Delta E}{k_B T}\right] \tag{6}$$

where N is the density-of-states in the band to which the carrier is emitted. ΔE is the level depth which is related to the activation

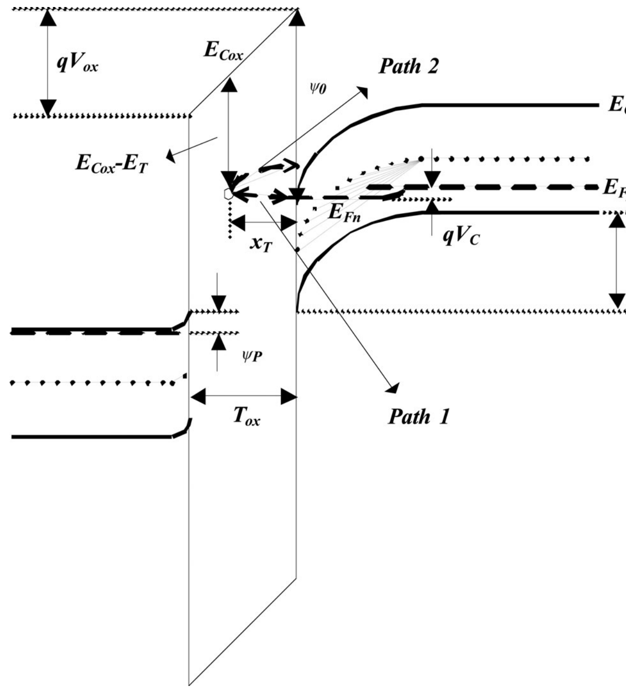

Fig. 2. Band diagram at the lateral position of traps.

energy for emission ΔE_{Eae}, and is given by $\Delta E = \Delta E_{Eae} - E_{\mathrm{oc}}$.

In a quantized inversion layer, the energy level can be derived using the triangular potential approximation [15]

$$E_{ij} = \left(\frac{\hbar^2}{2m_i}\right)^{1/3} \left[\frac{3}{2}\pi q F_S \left(j + \frac{3}{4}\right)\right]^{2/3}. \tag{7}$$

Here, F_S is the effective surface electric field and m_i is the density-of-states effective mass in the ith valley. The quantum mechanical surface layer effective density-of-states based on [16] is given by

$$N = N_{cqm} = \sum_i \sum_j D_i \cdot k_B T \cdot \exp\left(\frac{-E_{ij}}{k_B T}\right) \tag{8}$$

where D_i is the state density and given by $D_i = g_i m_i / \pi\hbar^2$. g_i is the degeneracy of the ith valley electrons.

Finally, the emission time constant τ_e can be calculated by the following equation:

$$\tau_e = \left(\frac{1}{\tau_{e\mathrm{SRH}}} + \frac{1}{\tau_{eT}}\right)^{-1}. \tag{9}$$

III. Results and Discussion

Two nMOSFET samples were measured to confirm the model equations. Sample 1 had an oxide thickness T_{ox} of 37 Å, a W/L of 0.12 μm/0.1 μm, and an effective gate length L_{eff} of about 100 nm. Sample 2 had a T_{ox} of 69 Å, a W/L of 0.16 μm/0.35 μm, and L_{eff} of about 276 nm. The samples had nearly the same threshold and flat-band voltages, about 530 mV and −0.98 V, respectively. The samples show two-level RTS noise characteristics in the drain current indicating that a trap

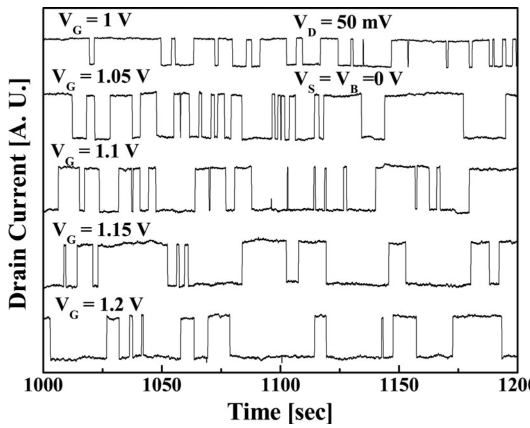

Fig. 3. τ_c and τ_e variation with respect to gate voltage. The RTS noise was measured from sample 2.

exists in the gate dielectric. Fig. 3 shows the gate bias dependence data for the RTS noise in each sample. In that figure, gate bias dependence of τ_c and τ_e can be observed.

For extraction of τ_c and τ_e for each gate bias from the measurement data, the first step is the generation of a quantized signal for a current fluctuation. The next step involves extracting capture and emission times. τ_c and τ_e are the means of the extracted capture and emission times. The methodological basis and the equations for calculation of the physical location and the associated energy levels from the extracted τ_c and τ_e data have been given by Lee *et al.* [17]. x_T and y_T were extracted from data obtained for each sample. The extracted x_Ts are 2.47 and 2.74 Å from the Si/SiO$_2$ surface, the y_Ts are 80.25 and 94.48 Å from the source, and the relative trap energy level to the gate oxide conduction band edge ($E_{C\,\mathrm{ox}}-E_T$) are 2.58 and 2.78 eV for samples 1 and 2, respectively.

To calculate τ_c, the meaning of E_{relax} needs to be clarified. After an electron is captured at a trap site, the oxide energy band and the lattice change slightly. For oxide relaxation, the captured electron has to excite onto an oxide conduction band and then be emitted to the channel. Naturally, the oxide lattice can be relaxed by a tunneling mechanism. However, E_{relax} is not the same as electron tunneling energy, or E_B. Instead, E_{relax} is the energy difference between the ground state of the trap and the oxide conduction band edge [11]; therefore, E_{relax} is the same as $E_{C\,\mathrm{ox}}-E_T$. For confirmation, data calculated using the model were compared with data measured from samples 1 and 2 (see Fig. 4). The $E_{C\,\mathrm{ox}}-E_T$ values for each sample that were extracted from the measurement data are used as the E_{relax} values in the model equation.

Figs. 5 and 6 show the verification results of the τ_e calculation. As shown in Figs. 5(a) and 6(a), there is a large difference between the calculated values of $\tau_{e\mathrm{SRH}}$ and the measured data in each of the samples. The calculated values of τ_{eT}, however, are more accurate. This means that the emission process along path 2 is more dominant than the emission along path 1. In strong inversion mode, the captured carrier is emitted to an opposite direction from the induced field and the energy for carrier emission is derived from thermal energy. The $\tau_{e\mathrm{SRH}}$ model mainly considers the trap energy with respect to Fermi energy

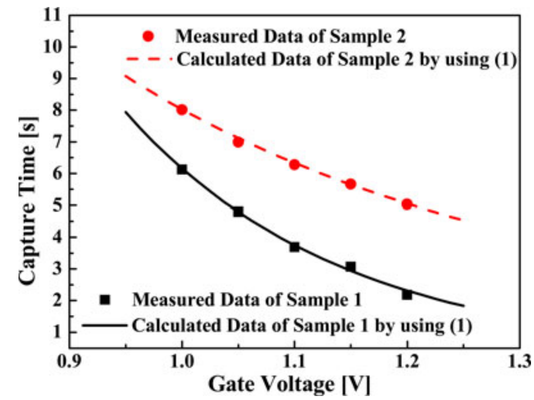

Fig. 4. Capture times versus gate voltage.

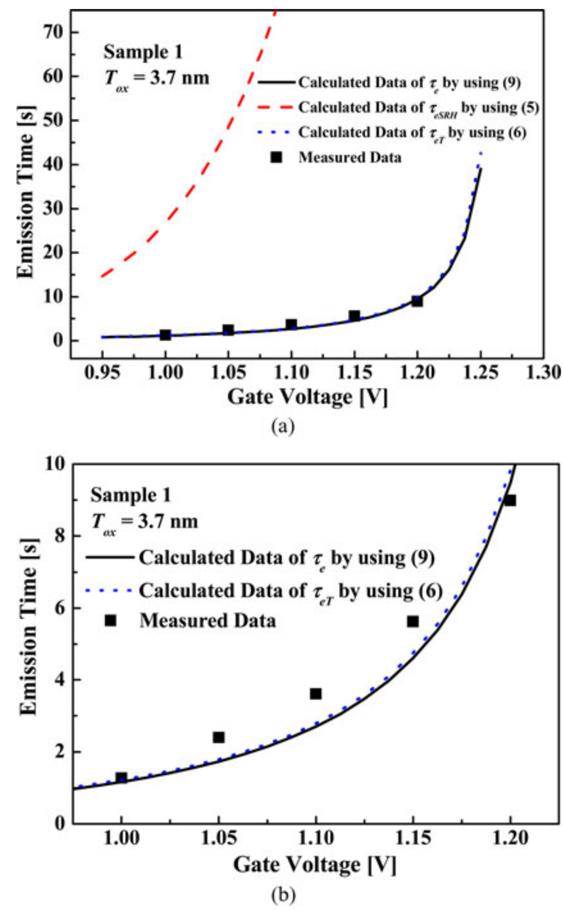

Fig. 5. Emission times versus gate voltage of sample 1. (a) Calculated dependences of each emission process. (b) Measurement data and calculated τ_e were shown in detail.

(E_F – E_T) and was derived from the occupancy of interfacial state that is near E_F with a trapped carrier. Thus, if the difference between E_F and E_T is increased, $\tau_{e\mathrm{SRH}}$ is increased in an exponential manner. However, the thermal energy increases the occupancy probability subsequently. On the other hand, the τ_{eT} model mainly considers the thermal energy gain of the captured carrier. Undoubtedly, the consideration of thermal energy and electric field are included in $\tau_{e\mathrm{SRH}}$ and τ_{eT} model, respectively. However, these are not the most crucial term in each

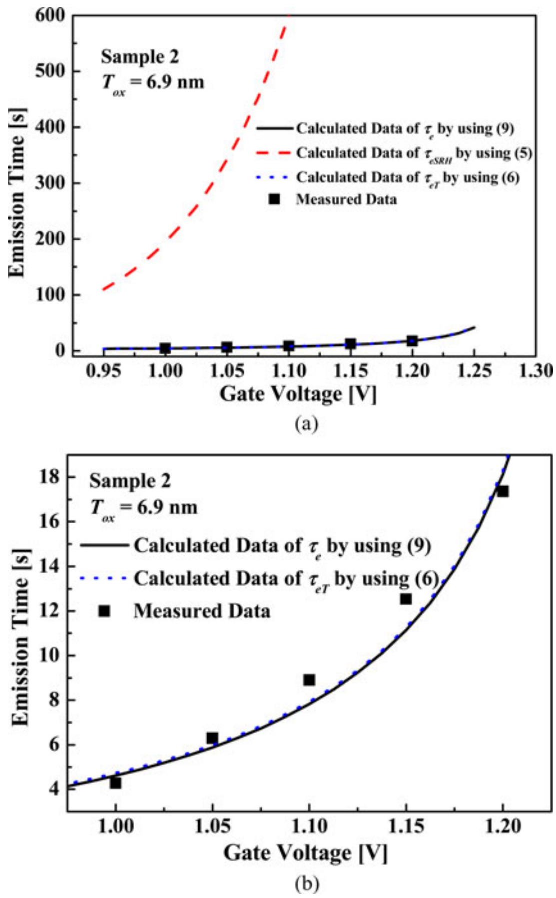

Fig. 6. Emission times versus gate voltage of sample 2. (a) Calculated dependences of each emission process. (b) Measurement data and calculated τ_e were shown in detail.

model. Therefore, the calculated values by the $\tau_{e\mathrm{SRH}}$ model are higher than measured data and τ_{eT} due to the rather simple mathematics which is invalid for full description of the thermal energy in the $\tau_{e\mathrm{SRH}}$ model. However, those two equations should be handled with care for the calculation of τ_e since both emission mechanisms can be performed indiscriminately. The dominant emission path is the path 2 in Fig. 2 and the driving mechanism is τ_{eT}. Therefore, the τ_{eT} model with the quantum effect is deemed to be the more important mechanism for the measured samples for this study. For more precise conformation, the region near the measured data is shown in Figs. 5(b) and 6(b).

IV. Conclusion

Here, a simple model for the capture process, based on SRH statistics including the "more fundamental" thermally activated σ_C model with the MPE capture theory, is introduced. The meaning of E_{relax}, which is the most important parameter of the τ_c model, is the energy difference between the ground state of the trap and the oxide conduction band edge. For the τ_e model, SRH statistics and a fundamental thermally activated emission time model with a quantum effect are considered. Both the SRH statistics and a fundamental thermally activated emission mechanism can be performed indiscriminately. Hence, consideration of both emission mechanisms is necessary for the analysis of the carrier emission process in the gate oxide of MOSFET devices.

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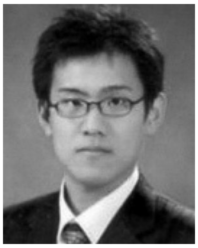

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